Covariance Matrix

Covariance : the covariance of two variables :

$$Cov(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$$

$$Cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Intuition :

$Cov(X,Y) > 0$, X and Y positive correlated

$Cov(X,Y) < 0$, X and Y negative correlated

$Cov(X,Y) = 0$, none linear correlation

random vector :

$$X = (X_1, X_2, \cdots, X_p)^T$$

$$\mu = \mathbb{E}[X]$$

$$= (\mathbb{E}[X_1], \mathbb{E}[X_2], \cdots, \mathbb{E}[X_p])^T$$

covariance matrix :

$$\Sigma = Cov(X)$$

$$= \mathbb{E}[(X-\mu)(X-\mu)^T]$$

$$\Sigma_{ij} = cov(X_i, X_j)$$

matrix form :

$$\Sigma = \begin{bmatrix} Var(X_1) & Cov(X_1,X_2) & \cdots & Cov(X_1,X_n) \\ Cov(X_2,X_1) & Var(X_2) & \cdots & Cov(X_2,X_n) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_n,X_1) & Cov(X_n,X_2) & \cdots & Var(X_n) \end{bmatrix}$$

Sample Covariance Matrix :

n sample

p variable

data matrix $X \in \mathbb{R}^{n \times p}$

$\bar{X} \in \mathbb{R}^{1 \times p}$

$1 \in \mathbb{R}^{n \times 1}$

Centering: $X_c = X - 1\bar{X}$

$$S = \frac{1}{n-1} X_c^T X_c$$

$$S_{ij} = \frac{1}{n-1} \sum_{k=1}^{n} (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

Independent and not correlated :

If $X_1, X_2, \ldots, X_n$ are independent,

$$\Sigma = \text{diag}(\text{Var}(X_1), \text{Var}(X_2), \ldots, \text{Var}(X_n))$$

But if $\Sigma = \text{diag}(\text{Var}(X_1), \text{Var}(X_2), \ldots, \text{Var}(X_n))$

$X_1, X_2, \ldots, X_n$ are not correlated

Correlation Value :

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

Shifts and Scales:

1. $Cov(aX+b, cY+d) = acCov(X, Y)$

2. W. A. X: matrix, $E(X) = \mu$. $Var(X) = \sigma^2$

$$\Sigma = Cov(X) = E[(X-\mu)(X-\mu)^T]$$

$$W = AX+b$$

$$Cov(W) = A\Sigma A^T$$

The eigenvector of Covariance Matrix

$$S\vec{u_i} = \lambda_i \vec{u_i}$$

λ_i: variance along the ith principal direction

u_i: principal direction themselves (orthogonal) ⇓ define uncorrelated directions

- $\lambda_i \uparrow$: larger variance in this direction
- $\lambda_i \downarrow$: smaller variance in this direction
- $\sum \lambda_i$: total variance in all direction